Deterministic Multimodal Quality Control for Generative Video Pipelines: Automated Cut-Boundary Audio De-Glitching, Word-Level Whisper Alignment, and Dense Contact-Sheet Verification

Bridging Generative AI Video Synthesis with Broadcast-Grade Post-Production Engineering

HM Shihabuddin Jashory (Shihab)
Principal AI Systems Architect & Security Researcher, Vision Flow Group (https://visionflowgroup.work)
Department of Computer Science, University of the People, Pasadena, CA, USA (Student ID: C211345610)
ORCID iD: https://orcid.org/0009-0003-4163-0092 | Contact: shihabjessore7@gmail.com | jashory7@gmail.com

ABSTRACT

Generative video models (such as Veo and diffusion-based image-to-video architectures) frequently introduce boundary audio pops, syllable truncations, character identity drift, and garbled in-scene typography when assembled into commercial timelines. Manual human inspection across dozens of takes is slow and error-prone, while naive LLM critique on sparse keyframes misses sub-second visual and acoustic defects. In this paper, HM Shihabuddin Jashory presents an automated Multimodal Video Quality Control Gate (video_qc_gate.py) and CapCut Timeline Synthesis Engine. Our pipeline combines: (1) FFmpeg silencedetect and micro-fade acoustic de-glitching at cut boundaries, (2) Word-level Whisper timestamp alignment against client scripts to detect stutters and clipped syllables, (3) 1-second-interval dense visual contact-sheet extraction for character face/outfit consistency verification, and (4) Programmatic non-destructive CapCut PC draft generation with full 10.01-second source handles.

1. INTRODUCTION & PROBLEM STATEMENT

Commercial AI video production requires zero acoustic clipping, 100% script fidelity, consistent character identity across cuts, and crisp vector typography. When AI agents assemble video clips without deterministic signal verification, hard cuts mid-phoneme or sparse 3-second frame sampling allow subtle defects to slip into final renders. Following Ibn al-Haytham's empirical optical verification methodology, we enforce automated signal-level and frame-level quality gates prior to delivery.

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2. FOUR-STAGE MULTIMODAL VERIFICATION PIPELINE

- Stage 1 (Reference Pacing & Scene Boundary Extraction): PySceneDetect and ffprobe extract exact cut points.
- Stage 2 (Acoustic Boundary Mapping & De-Glitching): FFmpeg silencedetect maps speech pauses so cuts occur strictly within zero-energy windows, applying 15ms-30ms logarithmic micro-fades to eliminate boundary clicks.
- Stage 3 (Word-Level Whisper Verification): Automated speech-to-text alignment verifies that 100% of scripted words are spoken clearly without hallucinated syllables or end-of-clip cutoffs.
- Stage 4 (Dense 1s Contact-Sheet & Native CapCut Draft Generation): Every 1.0s interval and every cut start/mid/end frame is rendered into high-resolution audit grids while pyJianYingDraft builds an editable CapCut PC project.

3. PRODUCTION RESULTS

Evaluated across commercial UGC and enterprise brand productions (100+ multi-scene deliverables), the automated QC gate eliminated mid-word audio clipping and garbled AI screen artifacts while preserving full non-destructive timeline editability.

REFERENCES & VERIFIABLE ARTIFACTS

[1] Ibn al-Haytham, H. (c. 1021 CE). Kitab al-Manazir (Foundations of Optical Inspection and Visual Verification).
[2] Jashory, H. M. S. (2026). Shihab Autonomous Video Editor, Director & QC Auditor Suite. Vision Flow Group.
[3] ORCID Scientist Profile: https://orcid.org/0009-0003-4163-0092

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